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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD1004-1 – COMPUTER NETWORKS

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is NOT a characteristic of an effective data communications system?
A) Delivery B) Cost-effectiveness
C) Accuracy D) Timeliness
2. In data communications, what does "jitter" refer to?
A) The speed of data transfer B) The amount of data lost during transmission
C) The variation in packet arrival time D) The delay in establishing a connection
3. A LAN typically connects devices within what geographical area?
A) A city B) A single office or building
C) A country D) The entire world
4. Which of the following is a characteristic of TCP?
A) Connectionless B) Connection-oriented
C) Unreliable D) Stateless
5. Which protocol is primarily used for email transmission?
A) FTP B) SMTP
C) HTTP D) UDP
6. What does the term "bandwidth-sensitive application" refer to?
A) Applications that require high bandwidth B) Applications that do not use bandwidth
C) Applications that can tolerate low bandwidth D) Applications that require no internet connection
7. Which protocol is used for secure communication over the web?
A) HTTP B) HTTPS
C) FTP D) SMTP
8. What does the term "internetwork" refer to?
A) A single network B) A local area network
C) A network of networks D) A wide area network
9. What does UDP stand for?
A) User Datagram Protocol B) User Data Packet
C) Universal Data Protocol D) Unreliable Datagram Protocol
10. In TCP, what is the purpose of the sequence number?
A) To identify the sender B) To order the segments of data
C) To ensure data integrity D) To manage connection states
11. What type of communication does the transport layer provide?
A) Physical communication B) Direct communication
C) Logical communication D) Encrypted communication

12. What is the primary role of a checksum in TCP and UDP?
 A) To encrypt data B) To manage flow control
 C) To detect errors in transmitted segments D) To establish connections
13. What is the main purpose of congestion control in TCP?
 A) To increase the speed of data transfer B) To ensure data is encrypted
 C) To prevent network overload D) To manage IP addresses
14. In a Go-Back-N protocol, what happens when a packet is lost?
 A) Only the lost packet is retransmitted B) The sender stops sending packets
 C) All packets after the lost packet are retransmitted D) The receiver discards all packets
15. What is the main advantage of using pipelining in TCP?
 A) It reduces the complexity of the protocol B) It simplifies error handling
 C) It increases the efficiency of data transfer D) It guarantees data integrity
16. Which of the following is a feature of the Selective Repeat protocol?
 A) It discards out-of-order packets B) It allows for individual retransmission of lost packets
 C) It acknowledges only the last packet received D) It requires a fixed window size
17. What is the primary purpose of a router in a network?
 A) Switch traffic B) Manage security
 C) Forward packets D) Provide DNS
18. Which layer of the OSI model is responsible for error detection?
 A) Physical B) Data Link
 C) Application D) Network
19. What is the maximum size of an IPv4 header without options?
 A) 16 bytes B) 20 bytes
 C) 32 bytes D) 40 bytes
20. What does ARP stand for?
 A) Address Resolution Protocol B) Access Routing Protocol
 C) Application Routing Protocol D) Address Recovery Protocol

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a diagram, describe the layers of the TCP/IP protocol suite and explain how they correspond to the OSI model.	8	L*2	1	1
	b) Demonstrate how the HTTP protocol facilitates communication between a web client and a web server and analyze its role in web browsing.	8	L3	2	2
2.	a) Compare different types of network topologies and analyze how the choice of topology impacts network performance and scalability.	8	L3	1	1
	b) Explain the architecture of an email system and analyze how email protocols (SMTP, POP3, IMAP) work together to ensure efficient message transmission and retrieval.	8	L2	2	2
3.	a) Classify different types of networks based on their scale and purpose and analyze how data communication varies across these network types.	8	L3	1	1
	b) Demonstrate how the Domain Name System (DNS) resolves domain names to IP addresses and analyze its role in ensuring efficient Internet communication.	8	L3	2	2

Unit – II

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| 4. a) | Explain the process of transport layer connectionless multiplexing and demultiplexing and analyze how they help in managing multiple network connections efficiently. | 8 | L2 | 3 | 1 |
| b) | Explain the working principles of the Go-Back-N (GBN) protocol with a suitable diagram. | 8 | L2 | 4 | 1 |
| 5. a) | With a neat diagram, illustrate the structure of a TCP segment and analyze the significance of its key fields in ensuring reliable data transmission. | 8 | L2 | 3 | 1 |
| b) | Illustrate the concept of the TCP congestion window and analyze how it dynamically adjusts to control network congestion. | 8 | L3 | 4 | 2 |
| 6. a) | Demonstrate the three-way handshake process in TCP and analyze its role in establishing a reliable connection between a client and a server. | 8 | L3 | 3 | 1 |
| b) | With a diagram, Illustrate the congestion scenario where two TCP connections share a single hop with infinite buffers, and analyze how it impacts network performance and fairness. | 8 | L3 | 4 | 2 |

Unit – III

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| 7. a) | Explain the format of an IPv4 packet and analyze the significance of its key fields in ensuring efficient packet delivery across networks. | 8 | L2 | 5 | 1 |
| b) | Consider a network with five routers (A, B, C, D, and E) connected as follows: <ul style="list-style-type: none"> • A is connected to B (cost = 2) and C (cost = 5). • B is connected to C (cost = 1) and D (cost = 3). • C is connected to D (cost = 2) and E (cost = 4). • D is connected to E (cost = 1). <p>Using Dijkstra's algorithm (Link State Routing), compute the shortest path from Router A to all other routers. Show the step-by-step execution of the algorithm and analyze how link state updates improve network routing efficiency.</p> | 8 | L3 | 5 | 2 |
| 8. a) | Explain the working of the Border Gateway Protocol (BGP) and analyze how it facilitates inter-domain routing between autonomous systems on the Internet. | 8 | L2 | 5 | 1 |
| b) | With an example, Illustrate the concept of Reverse Path Forwarding (RPF) and analyze how it helps in preventing routing loops and ensuring efficient multicast packet forwarding. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
